

AKSHANSH YADAV

Ph.D. Researcher | AI & FPGA Accelerators

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Research Profile

Ph.D. researcher in computer architecture and hardware–software co-design for AI inference. Develop lightweight token-pruning methods for Vision Transformers and FPGA accelerators that exploit reduced computation. Experience spans model evaluation, HLS architecture design, host–device integration, and analysis of accuracy, latency, power, and hardware resources.

Education

Indian Institute of Technology Jodhpur

July 2023 – Present

Integrated M.Tech.–Ph.D. in Computer Science and Engineering

M.Tech. awarded in 2025 | CGPA: **8.22** | Department rank: **2**

Research advisor: Dr. Palash Das

Dr. A.P.J. Abdul Kalam Technical University

2023

B.Tech. in Computer Science and Engineering

81.8%; First Division with Distinction

Sant Longowal Institute of Engineering and Technology

2018

Certificate Integrated Diploma in Mechanical Engineering; CGPA: 7.95

Research Experience

Vision Transformer Acceleration

Jan. 2024 – Present

IIT Jodhpur | Hardware–Software Co-Design | Advisor: Dr. Palash Das

- Develop early token-selection methods using entropy, cosine similarity, Kendall’s tau, and Pearson correlation; combine them with attention-based pruning in later encoder layers.
- Co-design **GateAttn-ViT**: achieve **30.8% fewer GOPs** on ViT-Base with **0.30% top-1 accuracy loss**. The ZCU104 accelerator reaches up to **20.9 FPS** and **5.39 FPS/W** at 250 MHz.
- Develop hybrid keypoint- and attention-based token selection with custom FPGA modules; the DATE 2025 design reduces execution time by **35.85%** against an unpruned baseline.
- Design correlation- and group-based pruning for **Prune Before You Attend**; reduce ViT-Base computation by approximately **39%** with around **1% accuracy loss**.

FPGA Inference Development on AMD Alveo U250

July 2025 – Present

IIT Jodhpur | Vitis HLS, Vivado, and XRT

- Develop inference kernels and host–device communication flows; optimize loops using pipelining, unrolling, and array partitioning to expose compute and memory parallelism.
- Analyze data movement, external-memory access, and LUT/BRAM/DSP utilization; evaluate latency and throughput trade-offs during accelerator development.

Hardware for Adversarial Robustness

May 2025 – Present

IIT Jodhpur | Efficient and Secure AI

- Contribute to hardware approaches for detecting and mitigating adversarial inputs, including **G-SHIELD** (IEEE iSES 2025) and **FreqPress** (submitted).

Technical Expertise

Languages	C, C++, Python, Verilog, HLS C/C++
FPGA & tools	AMD/Xilinx Vitis HLS, Vivado, Vitis IDE, XRT; ZCU104, Alveo U250, PYNQ
Architecture	Pipelining, loop unrolling, memory partitioning, cache/prefetch simulation
Machine learning	PyTorch, TensorFlow, Keras; ViTs, token pruning, adversarial robustness
Development	NumPy, Jupyter, Logisim, TCP sockets, LaTeX

Publications

Journal Articles

1. **A. Yadav** and P. Das. **Prune Before You Attend: Correlation-Driven Token Pruning for Hardware-Efficient ViT Acceleration.** *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2026. *Accepted*. DOI: 10.1109/TVLSI.2026.3737685.
2. **A. Yadav** and P. Das. **GateAttn-ViT: Entropy-Gated, Attention-Guided Token Pruning for Resource-Efficient Vision Transformer Acceleration on FPGAs.** *Journal of Systems Architecture*, vol. 177, article 103836, 2026. DOI: 10.1016/j.sysarc.2026.103836.

Conference Papers

3. **A. Yadav** and P. Das. **KTA-Attn ViT: A Hierarchical Token Pruning Framework for Accelerating ViTs.** *44th IEEE International Conference on Computer Design (ICCD)*, 2026. *Accepted*.
4. **A. Yadav** and P. Das. **CSAP-ViT: Cascade Similarity–Attention Pruning for Accelerating ViTs on FPGA.** *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026. *Accepted*.
5. D. Majumdar, **A. Yadav**, D. Raj, and P. Das. **G-SHIELD: Hardware Acceleration for Defending CNNs Against Adversarial Attacks.** *11th IEEE International Symposium on Smart Electronic Systems (iSES)*, 2025. *Accepted*.
6. **A. Yadav**, A. Goyal, and P. Das. **Hybrid Token Selector Based Accelerator for ViTs.** *Design, Automation & Test in Europe Conference (DATE)*, 2025. *Accepted*.

Submitted Manuscripts

- Anand Saxena, **Akshansh Yadav**, and Palash Das. **CRAFT: A Co-Designed Configurable Accelerator for Efficient DiT Inference via Adaptive Striding.** Submitted to the *Design, Automation & Test in Europe Conference (DATE)*, 2027.
- D. Majumdar, **A. Yadav**, and P. Das. **FreqPress: A Lightweight Hardware Accelerator for Robust Defense Against White-Box Adversarial Attacks.** Submitted to IEEE/ACM ESWEK, 2026.
- J. Debbarma, **A. Yadav**, R. Sharma, R. Kumar, P. G. Bahubalindrani, and P. Das. **Design and Implementation of an FPGA-Based Fruit Ripening Monitoring System Using E-Nose Technology.** Submitted to *IEEE Embedded Systems Letters*, 2026.

Selected Engineering Projects

GateAttn-ViT

GitHub repository

Vision Transformers | Token Pruning | FPGA Co-Design

Training-free entropy gating and attention-guided pruning for ViT inference; combines model evaluation with accelerator design on AMD ZCU104.

Last-Level Cache Prefetching in ChampSim

March 2024

C++ | Computer Architecture | Cache Simulation

Implement and integrate LLC prefetching policies in ChampSim to study cache behavior and compare prefetching strategies.

MIPS Processor Simulator and Assembly Translator

March 2023

Logisim | Python | MIPS Assembly

Build a MIPS instruction-execution simulator in Logisim and a Python translator that converts assembly instructions into compatible machine code.

Teaching & Industry Experience

Teaching Assistant, IIT Jodhpur

Fall 2023 – Present

Computer Architecture, Data Structures, Algorithm Design, and Introductory Computing

Conduct lab sessions; prepare exercises and course materials; evaluate student work.

Practical Training, Bharat Heavy Electricals Limited

Apr. 2019 – Apr. 2020

Electronics Assembly and Testing

Assist with component assembly and testing; gain exposure to solar-panel assemblies and lithium-ion battery production.

Honors & Academic Engagement

- **ACM ARCS 2026, IIT Hyderabad:** selected to attend and present research; received sponsored travel, accommodation, and hospitality.
- **IEEE VLSID 2025, Bengaluru:** awarded a competitive travel grant.
- **GATE 2023, Computer Science:** AIR 2337; score 532; marks 48.67/100.